1 Lightning occurs when charge builds up in the atmosphere, creating a potential difference between the ground and the atmosphere.

During a lightning strike there is an average current of 3.3×10^4 A for a time of 2.6×10^{-5} s.

(a) Calculate the charge transferred during the lightning strike.

charge = ... C [2]

(b) The potential difference between the ground and the atmosphere is 3.0×10^7 V.

Calculate the average power, in GW, transferred during the lightning strike.

power = ... GW [2]

(c) A lightning rod is attached to a tall building to conduct charge safely to the ground. The lightning rod is modelled as a uniform cylindrical copper cable of total length 95 m that runs from the ground to the top of the building, as shown in Fig. 3.1.

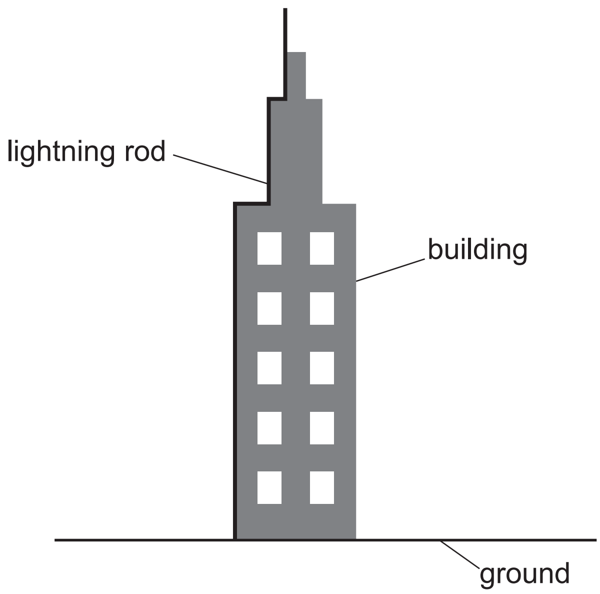

Fig. 3.1

(i) The resistance of the lightning rod is $9.6\,\Omega$.
The resistivity of copper is $1.7 \times 10^{-8}\,\Omega\,\text{m}$.

Determine the radius of the lightning rod.

radius = ... m [3]

(ii) The radius of the copper lightning rod is doubled with no change to its length.

State the effect of this change on the resistance of the lightning rod.

.... [1]

(d) A section of the lightning rod of length 0.12 m is removed for testing. A tensile stress of 1.9×10^{6} Pa is applied, as shown in Fig. 3.2.

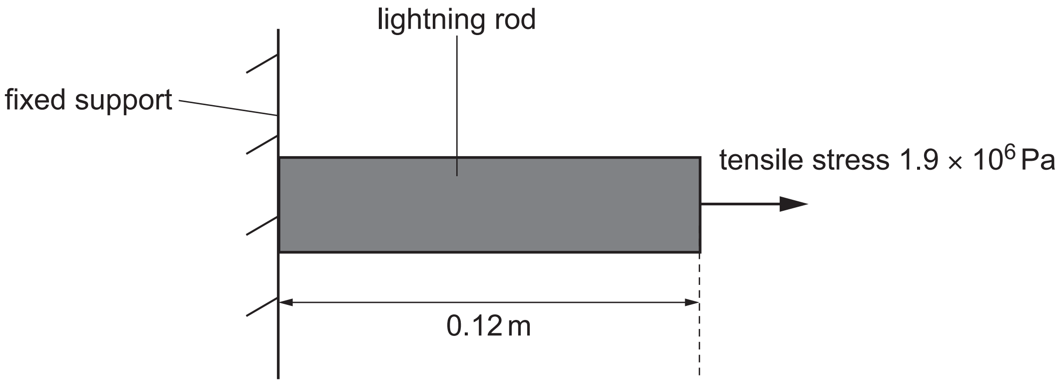

Fig. 3.2 (not to scale)

The section of the rod obeys Hooke's law. The Young modulus of copper is 1.3×10^{11} Pa.

Calculate the extension of the section.

extension = ... m [3]

[Total: 11]

2 **(a)** Define resistance.

....

.... [1]

(b) A cylindrical metal wire of length 2.4 m and cross-sectional area $8.0 \times 10^{-6}\,m^2$ has a resistance of $0.33\,\Omega$. There is a current in the wire of 4.7 A.

(i) Determine the resistivity of the metal from which the wire is made.

resistivity = ... $\Omega\,m$ [2]

(ii) Calculate the charge that passes through the wire in a time of 5.0 minutes.

charge = ... C [2]

(iii) The free electrons (charge carriers) in the wire have an average drift speed of $0.16\,mm\,s^{-1}$.

Determine the number density of charge carriers in the metal.

number density = ... m^{-3} [2]

(c) The wire in **(b)** may be considered to be a fixed resistor. It is connected in series with a thermistor to a battery that has negligible internal resistance.

(i) Use circuit symbols to complete Fig. 6.1 to show the circuit diagram of this arrangement.

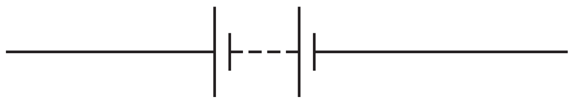

Fig. 6.1

[1]

(ii) Explain, without calculation, how the power dissipated in the wire changes as the temperature of the thermistor is increased.

....

....

....

.... [2]

[Total: 10]

3 **(a)** Define electric potential difference.

....

.... [1]

(b) A cell of electromotive force (e.m.f.) 1.8 V and internal resistance *r* is connected in parallel with a resistor of resistance 6.0 Ω and a filament lamp, as shown in Fig. 7.1.

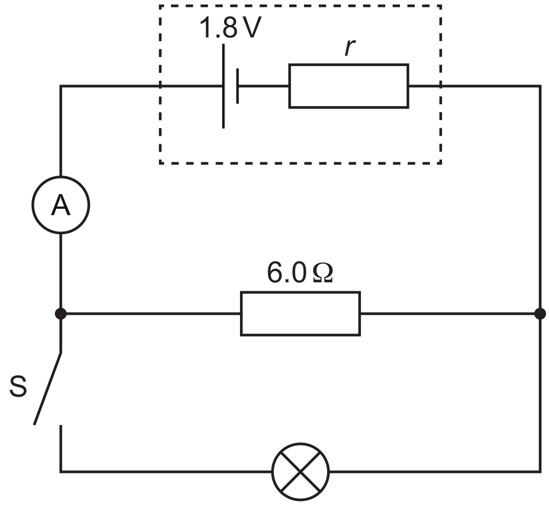

Fig. 7.1

The switch S is open. The ammeter reading is 0.25 A.

Determine the internal resistance *r* of the cell.

r = ... Ω [3]

(c) At time t_1 switch S in Fig. 7.1 is closed. Fig. 7.2 shows the variation with time t of the ammeter reading I.

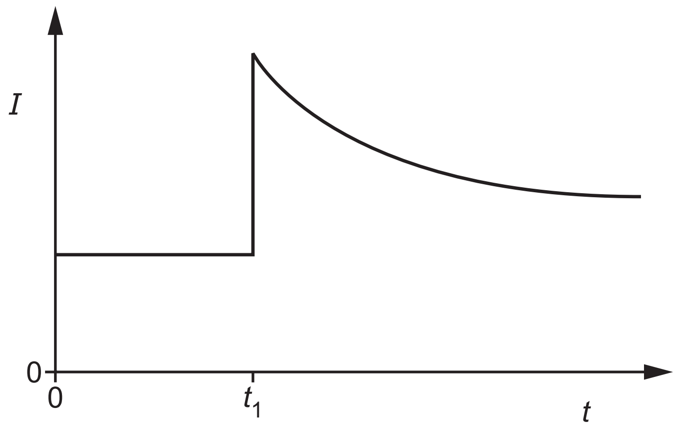

Fig. 7.2

(i) State whether the e.m.f. of the cell after t_1 is greater than, less than or the same as it was before t_1.

.... [1]

(ii) By considering the effect of the lamp on the total resistance of the circuit, explain the variation of the ammeter reading shown in Fig. 7.2.

....

....

....

....

....

.... [3]

[Total: 8]

4 **(a)** State Kirchhoff's first law.

....

.... [1]

(b) A cell with internal resistance r is connected to two resistors of resistances R_1 and R_2 as shown in Fig. 6.1.

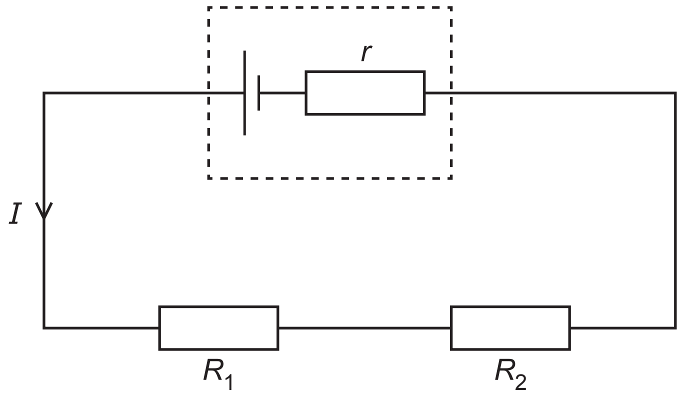

Fig. 6.1

The potential differences (p.d.s) across R_1 and R_2 are V_1 and V_2 respectively.
The terminal p.d. across the cell is V.
The current in the circuit is I.

Use Kirchhoff's laws to show that the total resistance R_T of the external circuit is given by

$$R_T = R_1 + R_2 .$$

[2]

(c) The electromotive force (e.m.f.) of the cell in Fig. 6.1 is 1.50 V.

The values of R_1 and R_2 are 10 Ω and 15 Ω respectively. The terminal p.d. of the cell is 1.35 V.

Calculate the internal resistance *r* of the cell.

r = ... Ω [3]

(d) A resistor of resistance R_3 is added to the circuit in Fig. 6.1, so that the circuit is as shown in Fig. 6.2.

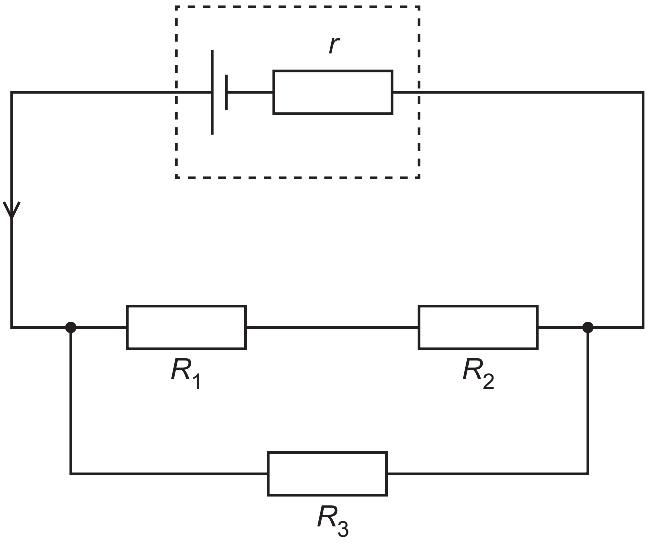

Fig. 6.2

State and explain the effect, if any, of this change on:

(i) the current in the cell

....

....

.... [2]

(ii) the terminal p.d. of the cell.

....

....

.... [2]

[Total: 10]

5 **(a)** State Kirchhoff's second law.

....

.... [1]

(b) A battery of electromotive force (e.m.f.) 9.0 V and negligible internal resistance is connected in series with a variable resistor X and a thermistor Y as shown in Fig. 5.1.

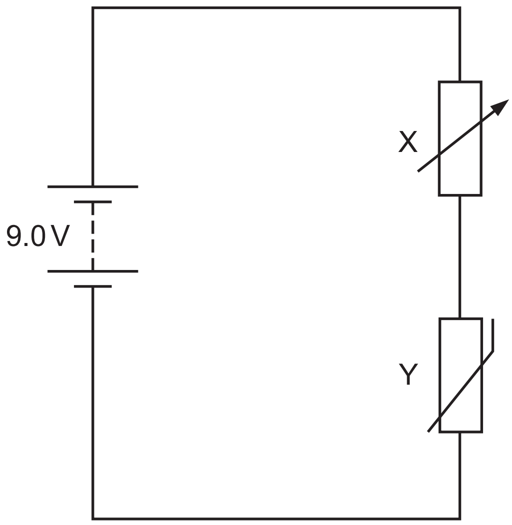

Fig. 5.1

Fig. 5.2 shows the relationship between temperature and resistance for the thermistor.

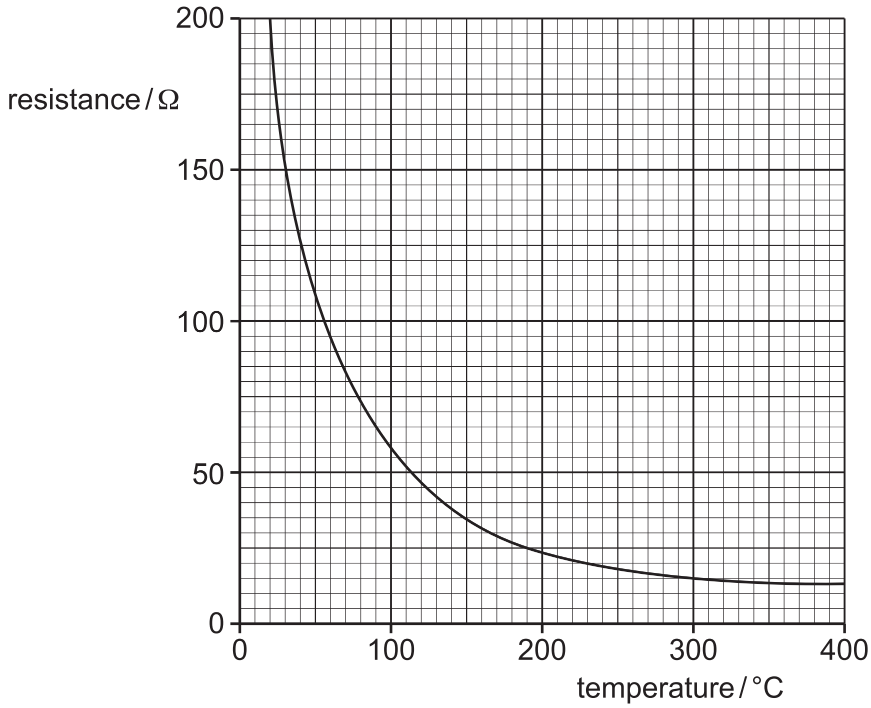

Fig. 5.2

(i) The current in the circuit is 1.1×10^{-2} A. The potential difference across Y is 4.0 V.

Calculate the resistance of X.

resistance = ... Ω [2]

(ii) The temperature of Y is changed to 190 °C. The resistance of X remains unchanged.

Determine the new potential difference across Y.

potential difference = ... V [3]

(iii) The resistance of X is increased. The temperature of Y remains at 190 °C.

By reference to the current in the circuit, state and explain the effect of this change, if any, on the potential difference **across Y**.

....

....

....

....

....

.... [3]

[Total: 9]

6 **(a)** **(i)** State Kirchhoff's second law.

....

....

.... [1]

(ii) State the conservation law that gives rise to Kirchhoff's second law.

.... [1]

(b) A circuit contains a cell of internal resistance r and two resistors of resistances R_1 and R_2, as shown in Fig. 5.1.

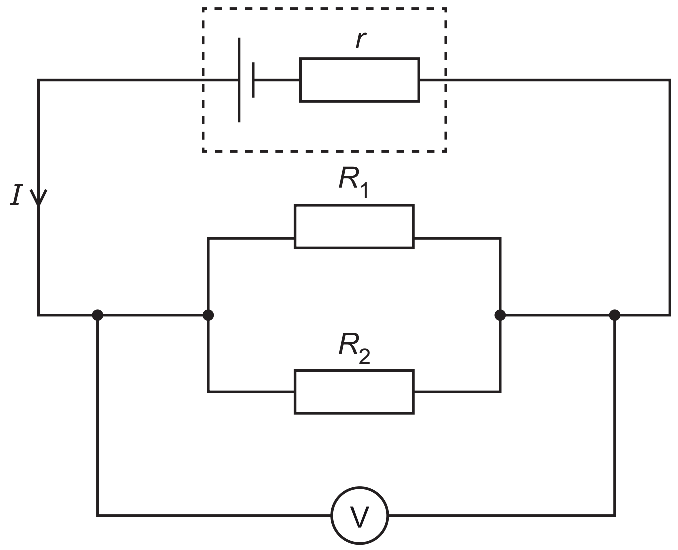

Fig. 5.1

The potential difference (p.d.) across the two resistors is V.

The current in the cell is I.

(i) Use Kirchhoff's laws to show that the total resistance R_T of the external circuit is given by

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}.$$

[2]

(ii) The electromotive force (e.m.f.) of the cell is 1.50 V.

When the values of R_1 and R_2 are 10 Ω and 15 Ω respectively, the p.d. measured by the voltmeter is 1.38 V.

Calculate the internal resistance r of the cell.

r = ... Ω [3]

(c) A third resistor is added in parallel with R_1 and R_2 in the circuit in Fig. 5.1.

State and explain the effect, if any, of this change on:

(i) the current in the cell

....

....

.... [2]

(ii) the p.d. measured by the voltmeter.

....

....

.... [2]

[Total: 11]

7 **(a)** Fig. 7.1 shows two resistors connected in series with a cell of electromotive force (e.m.f.) 1.50 V and internal resistance 0.28 Ω.

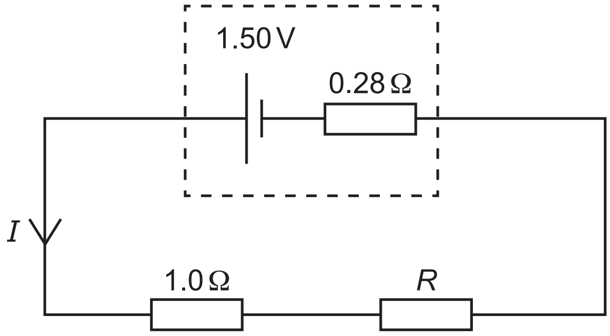

Fig. 7.1

One of the resistors has resistance 1.0 Ω. The other resistor has resistance *R*.
The terminal potential difference (p.d.) across the cell is 1.36 V.

(i) Show that the current I in the circuit is 0.50 A.

[2]

(ii) Calculate the combined resistance of the two resistors.

resistance = ... Ω [2]

(iii) Use your answer in **(a)(ii)** to determine resistance *R*.

R = ... Ω [1]

(b) The circuit in Fig. 7.1 is disconnected and the two resistors are reconnected to the cell, now in parallel with each other.

(i) On Fig. 7.2, complete the circuit diagram to show this arrangement.

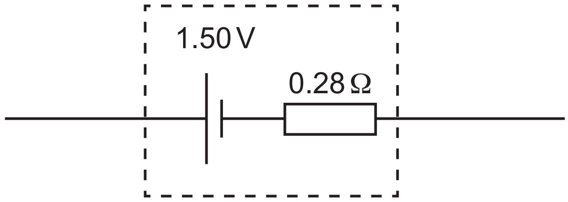

Fig. 7.2

[1]

(ii) Explain, without calculation, whether the terminal p.d. across the cell is now less than, equal to or greater than 1.36 V.

....

....

.... [2]

[Total: 8]

8 **(a)** Define electric potential difference (p.d.).

....

.... [1]

(b) A power supply, three resistors and a component X are connected in the circuit shown in Fig. 5.1.

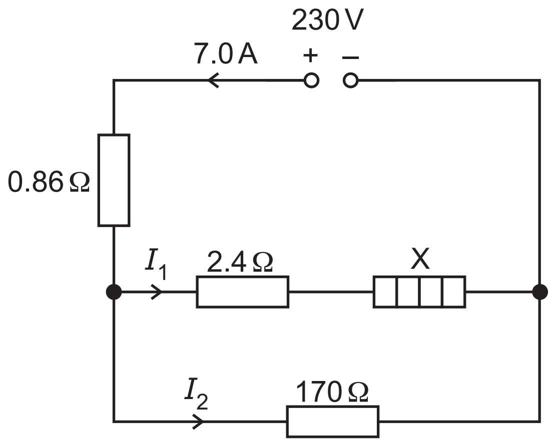

Fig. 5.1

The power supply has an electromotive force (e.m.f.) of 230 V and negligible internal resistance. The current in the power supply is 7.0 A.

(i) Identify component X.

.... [1]

(ii) Show that the p.d. across the resistor of resistance 0.86 Ω is 6.0 V.

[1]

(iii) Determine the current I_1.

I_1 = ... A [2]

(iv) Calculate the p.d. across component X.

p.d. = ... V [2]

(v) Calculate the power dissipated in component X.

power = ... W [2]

(vi) The purpose of the circuit is to provide power to component X.

Determine the percentage efficiency of the circuit.

efficiency = ...% [2]

(vii) The resistor of resistance 170 Ω is removed, leaving an open circuit in the lower branch of the circuit. There is no change to the resistance of component X.

State whether the current in the power supply increases, decreases or remains the same.

.... [1]

[Total: 12]